

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication	20250268769
Kind Code	A1
Publication Date	August 28, 2025
Inventor(s)	TAKEMOTO; Masaaki et al.

VEHICLE

Abstract

A vehicle includes front and rear wheels, a base portion, an occupant supporting portion, and a movement mechanism. The base portion includes a front frame supporting the front wheel and a rear frame supporting the rear wheel, and changes a wheelbase as a result of relative movement between the front frame and the rear frame. The occupant support portion includes a support surface. A vertical position of the support surface changes in conjunction with the relative movement between the front frame and the rear frame when the wheelbase changes. The movement mechanism moves the occupant support portion with respect to the base portion so as to change the vertical position of the support surface.

Inventors: TAKEMOTO; Masaaki (Toyokawa-shi, JP), HIRATA; Kazuyuki (Toyota-shi, JP)

Applicant: TOYOTA BOSHOKU KABUSHIKI KAISHA (Aichi-ken, JP)

Family ID: 1000008495906

Assignee: TOYOTA BOSHOKU KABUSHIKI KAISHA (Aichi-ken, JP)

Appl. No.: 19/052684

Filed: February 13, 2025

Foreign Application Priority Data

JP	2024-026340	Feb. 26, 2024
----	-------------	---------------

Publication Classification

Int. Cl.: A61G5/14 (20060101); A61G5/08 (20060101)

U.S. Cl.:

CPC A61G5/14 (20130101); A61G5/085 (20161101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2024-026340, filed on Feb. 26, 2024, the entire contents of which are incorporated herein by reference.

BACKGROUND

1. Field

[0002] The present disclosure relates to a vehicle.

2. Description of Related Art

[0003] Japanese Laid-Open Patent Publication No. 2010-167809 discloses a vehicle that includes a body equipped with a seat for an occupant, and front wheels and rear wheels supported by the body. In this vehicle, the rear wheels move in the longitudinal direction relative to the front wheels, thereby changing the wheelbase, which is the distance between the front wheels and the rear wheels. As a result, the orientation of the body changes, causing the center of gravity of the vehicle to change. When the wheelbase is at its minimum, the orientation of the body assumes an upright position relative to the ground. When the wheelbase is at its maximum, the orientation of the body tilts compared to the upright position.

[0004] In the vehicle disclosed in the above-described publication, the vertical position of the seat changes as the orientation of the body changes. Such changes in the vertical position of the seat can lead to issues such as reduced forward visibility for the occupant or diminished ease of getting on and off the vehicle. Therefore, there is room for improvement in improving the usability of the vehicle.

SUMMARY

[0005] This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

[0006] In one general aspect, a vehicle includes front and rear wheels, a base portion, an occupant support portion, and a movement mechanism. The base portion includes a front frame supporting the front wheel and a rear frame supporting the rear wheel. The base portion changes a wheelbase, which is a distance between the front wheel and the rear wheel, as a result of relative movement between the front frame and the rear frame. The occupant support portion includes a support surface configured to support an occupant from below. A vertical position of the support surface changes in conjunction with the relative movement between the front frame and the rear frame when the wheelbase changes. The movement mechanism couples the base portion and the occupant support portion and moves the occupant support portion with respect to the base portion so as to change the vertical position of the support surface.

[0007] Other features and aspects will be apparent from the following detailed description, the drawings, and the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a side view of a vehicle according to an embodiment.

[0009] FIG. 2 is a view of the vehicle as viewed in the direction of arrow A in FIG. 1.

[0010] FIG. 3 is a side view showing a state in which the wheelbase of the vehicle of FIG. 1 has changed.

[0011] FIG. 4 is a side view showing a state in which the seat of the vehicle of FIG. 1 has slid to a different position.

[0012] FIG. 5 is a side view showing a state in which the seat of the vehicle of FIG. 1 is inclined obliquely upward.

[0013] FIG. 6 is a side view showing a state in which the seat of the vehicle of FIG. 1 is inclined obliquely downward.

[0014] FIG. 7 is a side view showing a state in which the seat of the vehicle of FIG. 1 is separated from the base portion.

[0015] FIG. 8 is a side view showing a state in which the wheelbase has changed while the orientation of the seat of the vehicle of FIG. 1 is maintained.

[0016] FIG. 9 is a side view showing a vehicle according to a first modification.

[0017] FIG. 10 is a side view showing a state in which the wheelbase of the vehicle of FIG. 9 has changed.

[0018] Throughout the drawings and the detailed description, the same reference numerals refer to the same elements. The drawings may not be to scale, and the relative size, proportions, and depiction of elements in the drawings may be exaggerated for clarity, illustration, and convenience.

DETAILED DESCRIPTION

[0019] This description provides a comprehensive understanding of the methods, apparatuses, and/or systems described. Modifications and equivalents of the methods, apparatuses, and/or systems described are apparent to one of ordinary skill in the art. Sequences of operations are exemplary, and may be changed as apparent to one of ordinary skill in the art, with the exception of operations necessarily occurring in a certain order. Descriptions of functions and constructions that are well known to one of ordinary skill in the art may be omitted.

[0020] Exemplary embodiments may have different forms, and are not limited to the examples described. However, the examples described are thorough and complete, and convey the full scope of the disclosure to one of ordinary skill in the art.

[0021] In this specification, “at least one of A and B” should be understood to mean “only A, only B, or both A and B.”

[0022] A vehicle 10 according to an embodiment will now be described with reference to FIGS. 1 to 8.

[0023] In the following description, the ‘forward’ and ‘rearward’ in the longitudinal direction of the vehicle 10 shown in FIG. 1 will simply be referred to as ‘forward’ and ‘rearward’, respectively. Also, the ‘upward’ and ‘downward’ in the vertical direction of the vehicle 10 will simply be referred to as the ‘upward’ and ‘downward’, respectively.

Vehicle 10

[0024] As shown in FIGS. 1 and 2, the vehicle 10 is, for example, a compact vehicle designed for single-person use. The vehicle 10 includes a pair of front wheels 20, a pair of rear wheels 30, and a base portion 40 that supports the front wheels 20 and the rear wheels 30. The vehicle 10 further includes a seat 50 for an occupant and a movement mechanism 60 that couples the base portion 40 to the seat 50. In FIG. 2, the seat 50 is omitted from the illustration.

[0025] The front wheels 20 are spaced apart from each other in the vehicle width direction. The rear wheels 30 are disposed behind the front wheels 20 and spaced apart from each other in the vehicle width direction. Each of the rear wheels 30 is a drive wheel driven by a motor (not shown). For example, the motor is driven in response to an operation of an operation portion (not shown) by the occupant.

Base Portion 40

[0026] The base portion 40 includes a front frame 41, a rear frame 44, and a support rod 47. The front frame 41 supports the front wheels 20. The rear frame 44 supports the rear wheels 30. The support rod 47 extends in the vehicle width direction and supports the front frame 41 and the rear frame 44 such that the front frame 41 and the rear frame 44 are rotatable relative to each other.

[0027] As shown in FIG. 2, the front frame **41** includes a pair of front side-frame members **42** and a pair of coupling rods **43**, which couple the front side-frame members **42** to each other.

[0028] The front side-frame members **42** are spaced apart from each other in the vehicle width direction. Each of the front side-frame members **42** has the shape of an elongated plate. A front end portion of each front side-frame member **42** rotatably supports the corresponding front wheel **20** via a front shaft **21**. A rear end portion of each front side-frame member **42** is rotatably supported by the support rod **47**.

[0029] The coupling rods **43** extend in the vehicle width direction and are spaced apart from each other in the longitudinal direction of the front side-frame members **42**. Each coupling rod **43** is rotatably supported by the front side-frame members **42**. Each of the coupling rods **43** is rotated in forward and reverse directions around the axis of the coupling rod **43** by the driving force of an electric first driving device **D1**. The two coupling rods **43** are respectively driven by the two first driving devices **D1**.

[0030] The rear frame **44** includes a pair of rear side-frame members **45** and a coupling frame **46** that couples the rear side-frame members **45** to each other.

[0031] The rear side-frame members **45** are spaced apart from each other in the vehicle width direction. Each of the rear side-frame members **45** has the shape of an elongated plate. A front end portion of each rear side-frame member **45** is non-rotatably supported by the support rod **47**. A rear end portion of each rear side-frame member **45** rotatably supports the corresponding rear wheel **30** via a rear shaft **31**.

[0032] The support rod **47** is rotated in forward and reverse directions around the axis of the support rod **47** by the driving force of an electric second driving device **D2**.

[0033] As shown in FIGS. 1 and 3, when the support rod **47** rotates, the front frame **41** and the rear frame **44** rotate relative to each other. The base portion **40** changes a wheelbase **H**, which is the distance between the front wheels **20** and the rear wheels **30**, as a result of relative rotation between the front frame **41** and the rear frame **44** about the support rod **47**.

[0034] The base portion **40** includes a lock mechanism (not shown) that locks the front frame **41** and the rear frame **44** at any desired position when the front frame **41** and the rear frame **44** are rotated relative to each other. Accordingly, the vehicle **10** is capable of traveling with the changed wheelbase **H**. In other words, the vehicle **10** is capable of traveling in multiple configurations in which the wheelbase **H** differs from one configuration to another.

[0035] The vehicle **10** changes the wheelbase **H** in accordance with the vehicle speed, for example. When traveling at low speeds, the vehicle **10** reduces the wheelbase **H**, thereby moving the center of gravity of the vehicle **10** upward. Conversely, when traveling at high speeds, the vehicle **10** increases the wheelbase **H**, thereby moving the center of gravity of the vehicle **10** downward.

Seat 50

[0036] The seat **50** includes a seat portion **51** and a backrest portion **52**. The seat portion **51** has a support surface **51a** configured to support the buttocks of the occupant from below. The seat **50** is an example of an “occupant support portion”.

[0037] The vertical position of the support surface **51a** of the seat **50** changes in conjunction with relative rotation between the front frame **41** and the rear frame **44** when the wheelbase **H** changes. Specifically, the orientation of the front frame **41** changes in conjunction with the relative rotation, so that the orientation of the seat **50** changes. Accordingly, the vertical position of at least a part of the support surface **51a** differs between the state before the change in the wheelbase **H** and the state after the change.

Movement Mechanism 60

[0038] As shown in FIGS. 1 and 2, the movement mechanism **60** includes a pair of movable rails **61**, a pair of support rails **62**, and a pair of link mechanisms **70**. The movable rails **61** are spaced apart from each other in the vehicle width direction. The support rails **62** are spaced apart from each other in the vehicle width direction. The link mechanisms **70** are spaced apart from each other

in the vehicle width direction. Each of the movable rails **61** and the support rails **62** has an elongated shape extending along the front side-frame members **42**. Each movable rail **61** is attached to the seat **50** with a bracket (not shown). Each support rail **62** slidably supports the corresponding movable rail **61**. Each support rail **62** is supported by the corresponding front side-frame member **42** via the corresponding link mechanism **70** such that the support rail **62** is rotatable in forward and reverse directions about an axis extending in the vehicle width direction. [0039] As shown in FIG. **4**, each of the movable rails **61** is configured to be slidable along the support rail **62** between a first position and a second position, which is higher than the first position.

[0040] As shown in FIG. **2**, the movable rails **61** are slid along the support rails **62** by the driving force of a single electric third driving device **D3**, for example. The movable rails **61** slide in conjunction with each other.

[0041] As shown in FIGS. **1** and **2**, two brackets **63** are attached to the lower surface of each support rail **62**. The brackets **63** are spaced apart from each other in the longitudinal direction of the support rail **62**.

[0042] Each link mechanism **70** includes a front coupling portion **71A** and a rear coupling portion **71B**. The front coupling portion **71A** and the rear coupling portion **71B** are spaced apart from each other in the longitudinal direction of the corresponding front side-frame member **42**. The front coupling portion **71A** and the rear coupling portion **71B** have the same configuration, for example. Hereinafter, the configuration of the rear coupling portion **71B** will be described, and the description of the configuration of the front coupling portion **71A** may be omitted.

[0043] The rear coupling portion **71B** includes a first link **72**, a second link **73**, and a rotation shaft **74**. The first link **72** is non-rotatably coupled to a distal end of the corresponding coupling rod **43** that protrudes outward in the vehicle width direction from the corresponding front side-frame member **42**. The second link **73** is rotatably supported by the bracket **63** via a pin **75** extending in the vehicle width direction. The rotation shaft **74** extends in the vehicle width direction and couples the first link **72** and the second link **73** to each other such that the first link **72** and the second link **73** are rotatable relative to each other. The rotation shaft **74** synchronizes rotations of the first link **72** and the second link **73** such that the rotation angles of the first link **72** and the second link **73** with respect to the rotation shaft **74** are the same.

[0044] The first link **72** rotates about the coupling rod **43** in conjunction with rotation of the coupling rod **43**. The second link **73** rotates around the pin **75** in conjunction with the rotation of the first link **72**. At this time, the first link **72** and the second link **73** rotate in opposite directions by the same rotation angles with respect to the rotation shaft **74**.

[0045] As shown in FIGS. **1** and **5**, when the first links **72** and the second links **73** of the front coupling portions **71A** rotate, the support rails **62** rotate in the forward direction or the reverse direction about the central axis of the pins **75** of the rear coupling portions **71B**. This tilts the support rails **62** with respect to the front frame **41**. Specifically, the support rails **62** are inclined with respect to the front frame **41** such that the distance between the front ends of the support rails **62** and the front frame **41** is larger than the distance between the rear ends of the support rails **62** and the front frame **41**. As a result, the orientation of the seat **50** with respect to the base portion **40** changes so as to face obliquely upward.

[0046] As shown in FIGS. **1** and **6**, when the first links **72** and the second links **73** of the rear coupling portions **71B** rotate, the support rails **62** rotate in the forward direction or the reverse direction about the central axis of the pins **75** of the front coupling portions **71A**. This tilts the support rails **62** with respect to the front frame **41**. Specifically, the support rails **62** are inclined with respect to the front frame **41** such that the distance between the rear ends of the support rails **62** and the front frame **41** is larger than the distance between the front ends of the support rails **62** and the front frame **41**. As a result, the orientation of the seat **50** with respect to the base portion **40** changes so as to face obliquely downward.

[0047] As shown in FIG. 7, the link mechanisms **70** adjust the rotation angle of the coupling rods **43** to move the support rails **62** to multiple positions at which the orientation of the support rails **62** with respect to the base portion **40** is consistent and the distance between the support rails **62** and base portion **40** is different from one position to another. Specifically, when the two coupling rods **43** are simultaneously or alternately driven, the entire support rails **62** move toward or away from the base portion **40**.

[0048] When the two coupling rods **43** are driven simultaneously, the support rails **62** move toward or away from the base portion **40** while maintaining the orientation with respect to the base portion **40**. Therefore, the seat **50** moves toward or away from the base portion **40** with the orientation of the seat **50** maintained with respect to the base portion **40**.

[0049] When the two coupling rods **43** are alternately driven, the support rails **62** are inclined with respect to the base portion **40** with the front coupling portions **71A** or the rear coupling portions **71B** as the starting points. Then, the support rails **62** are inclined with respect to the base portion **40** with the other of the front coupling portions **71A** and the rear coupling portions **71B** as the starting points. Therefore, the seat **50** moves toward or away from the base portion **40** while changing the orientation with respect to the base portion **40**.

[0050] Further, as shown in FIGS. 1 and 8, the link mechanisms **70** are capable of adjusting the rotation angles of the two coupling rods **43** to maintain the orientation of the seat **50** relative to the direction of gravitational force consistent between the state before and the state after the change in the wheelbase H. At this time, the vertical position of the support surface **51a** of the seat **50** is also kept consistent between the state before and the state after the change in the wheelbase H.

[0051] As described above, the movement mechanism **60** can move the seat **50** with respect to the base portion **40** so as to change the vertical position of the support surface **51a** by driving the movable rails **61** and/or the link mechanisms **70**. Accordingly, the vehicle **10** is capable of changing the orientation of the seat **50** to any desired orientation while the vehicle **10** is traveling or is in a stopped state.

Operation of the Present Embodiment

[0052] As shown in FIGS. 1 and 3, relative movement between the front frame **41** and the rear frame **44** changes the wheelbase H, which in turn changes the vertical position of the support surface **51a** of the seat **50** changes. As a result, the center of gravity of the vehicle **10** is altered. Accordingly, the center of gravity of the vehicle **10** can be adjusted to improve the stability of the vehicle **10** during operation by changing the wheelbase H in accordance with the vehicle speed.

[0053] Further, as shown in FIG. 4, even when the vertical position of the support surface **51a** of the seat **50** is changed when the wheelbase H is changed, the vertical position of the support surface **51a** can be changed by the movement mechanism **60** without changing the wheelbase H.

Advantages of the Present Embodiment

[0054] (1) The base portion **40** changes the wheelbase H as a result of relative movement between the front frame **41** and the rear frame **44**. The vertical position of the support surface **51a** of the seat **50** changes in conjunction with relative movement between the front frame **41** and the rear frame **44** when the wheelbase H changes. The movement mechanism **60** moves the seat **50** with respect to the base portion **40** so as to change the vertical position of the support surface **51a**.

[0055] This configuration, which achieves the above-described operation, improves poor forward visibility from the vehicle **10** for the occupant and the ease of getting on and off the vehicle **10**. The usability of the vehicle **10** is improved, accordingly.

[0056] (2) The wheelbase H changes as a result of relative rotation between the front frame **41** and the rear frame **44** about the support rod **47**.

[0057] With this configuration, the wheelbase H can be easily changed by rotating the front frame **41** and the rear frame **44** relative to each other. This allows the center of gravity of the vehicle **10** to be readily adjusted.

[0058] (3) Each of the movable rails **61** is configured to be slidable along the support rail **62**

between a first position and a second position, which is higher than the first position.

[0059] With this configuration, when the movable rails **61** slide along the support rails **62**, the seat **50**, which is fixed to the movable rail **61**, slides between the first position and the second position. Accordingly, the vertical position of the support surface **51a** of the seat **50** can be readily changed without changing the wheelbase **H**.

[0060] Further, as indicated by the solid lines in FIG. **4**, moving the seat **50** to the first position improves the ease of getting on and off the vehicle **10**. Further, as indicated by the long-dash double-short-dash lines in FIG. **4**, moving the seat **50** to the second position raises the occupant's eye level, thereby improving the forward visibility from the vehicle **10** for the occupant. In addition, the visibility of the occupant as perceived by others is improved.

[0061] (4) The movement mechanism **60** includes the link mechanisms **70**, via which the base portion **40** supports the support rails **62** such that the support rails **62** are rotatable in forward and reverse directions about the axis extending in the vehicle width direction.

[0062] This configuration inclines the support rails **62** with respect to the base portion **40** about the axis extending in the vehicle width direction, allowing the seat **50** to be inclined with respect to the base portion **40**. This allows the posture of the occupant to be changed without changing the wheelbase **H**.

[0063] Further, as shown in FIG. **5**, inclining the seat **50** obliquely upward improves the riding comfort of the occupant. Additionally, as shown in FIG. **6**, inclining the seat **50** obliquely downward improves the ease of getting on and off the vehicle **10**.

[0064] (5) The link mechanisms **70** move the support rails **62** to multiple positions at which the orientation of the support rails **62** with respect to the base portion **40** is consistent and the distance between the support rails **62** and base portion **40** is different from one position to another.

[0065] This configuration allows the distance of the seat **50** from the base portion **40** to be changed without changing the orientation of the seat **50** with respect to the base portion **40**. Therefore, the center of gravity of the vehicle **10** can be adjusted without changing the wheelbase **H**, while maintaining the posture of the occupant. Thus, the stability of the vehicle **10** during operation is improved by adjusting the center of gravity of the vehicle **10** in accordance with the vehicle speed.

Modifications

[0066] The above-described embodiment may be modified as follows. The above-described embodiment and the following modifications can be combined as long as the combined modifications remain technically consistent with each other.

[0067] The link mechanisms **70** do not necessarily need to move the entire support rails **62** toward or away from the base portion **40**. The link mechanisms **70** may support the support rails **62** in a manner movable with respect to the base portion **40** such that the rear ends of the support rails **62** move upward with the front ends as the centers of rotation, or such that the front end portions of the support rails **62** move upward with the rear ends as the centers of rotation.

[0068] The link mechanisms **70** are not limited to the configuration of the above-described embodiment, and may have various configurations as long as the link mechanisms **70** can move the support rails **62** with respect to the base portion **40**.

[0069] The movement mechanism **60** does not necessarily need to include the link mechanisms **70** as long as the movement mechanism **60** includes the support rails **62** and the movable rails **61**. In this case, for example, the support rails **62** are directly or indirectly fixed to the front frame **41**. Even with such a configuration, it is possible to achieve an advantage similar to the advantage (3).

[0070] The movement mechanism **60** does not necessarily need to include the movable rails **61**, the support rails **62**, and the link mechanisms **70**. The movement mechanism **60** may couple the base portion **40** and the seat **50** to each other and rotate the seat **50** in forward and reverse directions about an axis extending in the vehicle width direction. Even in this case, since the vertical position of the support surface **51a** changes as a result of rotation of the seat **50**, it is possible to achieve an advantage similar to the advantage (1).

[0071] The movement mechanism **60** may couple the rear frame **44** and the seat **50** to each other. In other words, the support rails **62** may be supported by the rear frame **44** with the link mechanisms **70**.

[0072] As shown in FIGS. **9** and **10**, the base portion **40** does not necessarily need to include the support rod **47**. In this case, for example, the rear frame **44** may be configured to extend and retract relative to the front frame **41**. In this modification, the rear frame **44** is slidably housed within an accommodating recess **41a** formed inside the front frame **41**. Accordingly, the wheelbase **H** changes as the rear frame **44** extends or retracts relative to the front frame **41**.

[0073] Instead of the seat **50**, the vehicle **10** may include an occupant support portion that supports an occupant in a standing position from below, or may include an occupant support portion that supports an occupant in a lying position from below.

[0074] Various changes in form and details may be made to the examples above without departing from the spirit and scope of the claims and their equivalents. The examples are for the sake of description only, and not for purposes of limitation. Descriptions of features in each example are to be considered as being applicable to similar features or aspects in other examples. Suitable results may be achieved if sequences are performed in a different order, and/or if components in a described system, architecture, device, or circuit are combined differently, and/or replaced or supplemented by other components or their equivalents. The scope of the disclosure is not defined by the detailed description, but by the claims and their equivalents. All variations within the scope of the claims and their equivalents are included in the disclosure.

Claims

1. A vehicle, comprising: front and rear wheels; a base portion that includes a front frame supporting the front wheel and a rear frame supporting the rear wheel, the base portion changing a wheelbase, which is a distance between the front wheel and the rear wheel, as a result of relative movement between the front frame and the rear frame; an occupant support portion including a support surface configured to support an occupant from below, a vertical position of the support surface changing in conjunction with the relative movement between the front frame and the rear frame when the wheelbase changes; and a movement mechanism that couples the base portion and the occupant support portion and moves the occupant support portion with respect to the base portion so as to change the vertical position of the support surface.
 2. The vehicle according to claim 1, wherein the base portion includes a support shaft that extends in a vehicle width direction and supports the front frame and the rear frame such that the front frame and the rear frame are rotatable relative to each other, and the wheelbase changes as a result of relative rotation between the front frame and the rear frame about the support shaft.
 3. The vehicle according to claim 1, wherein the movement mechanism includes: a movable rail attached to the occupant support portion; and a support rail that slidably supports the movable rail, and the movable rail is configured to be slidable along the support rail between a first position and a second position, the second position being higher than the first position.
 4. The vehicle according to claim 3, wherein the movement mechanism includes a link mechanism via which the base portion supports the support rail such that the support rail is rotatable in forward and reverse directions about an axis extending in a vehicle width direction.
 5. The vehicle according to claim 4, wherein the link mechanism moves the support rail to multiple positions at which an orientation of the support rail with respect to the base portion is consistent and a distance between the support rail and the base portion is different from one position to another.
-